

AI Data Analyst Report

Dataset Overview

File Name: ecommerce_data.csv

Rows: 20

Columns: 13

Business Type: E-commerce and Retail

Business Summary

A high-quality transactional dataset of 20 orders and 13 variables with no missing or duplicate records. It captures critical e-commerce metrics such as gross revenue, order volume, and geographic distribution, enabling stakeholders to assess operational health, analyze category performance, and optimize logistics.

Key Insights

- **High Customer and Product Fragmentation:** Every single transaction (20 out of 20) represents a unique customer, a unique product, and a unique shipping city, indicating a lack of repeat purchases or product concentration during this 10-day period.
- **Price-Quantity Elasticity:** There is a strong negative correlation of -0.62 between unit price and quantity, demonstrating that customers limit their purchases of high-priced items to lower volumes while purchasing cheaper items in higher quantities.
- **Revenue Driven by Unit Price:** Total price has a very high positive correlation with unit price (0.91) but a negative correlation with quantity (-0.36), proving that revenue growth is driven by high-ticket item sales rather than bulk volume purchases.
- **Logistical Complexity:** Shipments are highly decentralized across 16 unique countries, with India (3 orders), USA (2 orders), and South Korea (2 orders) being the only markets with multiple transactions, indicating high operational overhead for a small order volume.
- **Order Fulfillment Risk:** Out of 20 orders, 2 have been cancelled (10%) and 1 remains pending (5%), representing a combined 15% non-delivery rate that could indicate payment friction or inventory availability issues.
- **Traditional Payment Dominance:** Credit Cards (8 orders) and PayPal (5 orders) are the preferred transaction methods, accounting for 65% of total transactions, while digital wallets like UPI (3 orders) remain a minority choice.
- **Metadata Profiling Anomaly:** The data profiling output contains a system error where numeric fields (order_id, quantity, unit_price, total_price) are erroneously parsed and categorized under datetime_columns with Unix epoch timestamps.

Recommendations

- **Implement Strategic Product Bundling:** Create curated product bundles of low-cost items to incentivize larger basket sizes and combat the negative correlation between quantity and total price.
- **Optimize International Shipping Hubs:** Establish localized distribution channels or partnerships in India, USA, and South Korea to streamline fulfillment costs in the business's most active geographic markets.
- **Analyze Cancellation Root Causes:** Investigate the 2 cancelled orders to determine if cart abandonment, local payment gateway failures, or unexpected shipping costs/times triggered the cancellations.
- **Enhance Regional Checkout Options:** Leverage local payment methods (such as promoting UPI in India) to increase conversion rates and decrease the friction that lead to pending or cancelled orders.
- **Resolve Data Pipeline Errors:** Coordinate with the data engineering team to correct the metadata schema mapping so that numeric metrics are not incorrectly classified as datetime variables in automated analytics pipelines.

Forecast

Forecast module will be implemented in the next phase.

Generated by AI Data Analyst